

Sales Forecasting & Demand Intelligence System

Business Report for the Head of Supply Chain and Chief Financial Officer (CFO)

Summary

This report presents the findings of a sales forecasting and demand intelligence analysis conducted using four years of retail sales data. The objective was to understand historical sales patterns, identify unusual sales events, forecast future demand, and recommend inventory strategies that improve business performance. The analysis shows that sales have grown steadily over time with clear seasonal peaks during the final months of each year. Three forecasting models were evaluated, with XGBoost providing the highest prediction accuracy. The report also identifies key sales anomalies, highlights future demand by product category and region, and provides practical recommendations to improve inventory planning, reduce stock shortages, and support better financial decision-making.

Key Findings from Exploratory Data Analysis and Forecasting

1. Technology is the Highest Revenue-Generating Category

Among all product categories, Technology generated the highest total sales revenue, followed by Furniture and Office Supplies. This indicates that Technology products continue to be the company's strongest revenue driver.

2. Central Region Shows the Most Stable Sales

Although the West and East regions generated higher sales in several years, the Central region demonstrated the most consistent sales performance across the four-year period. Stable demand makes inventory planning more predictable and reduces forecasting uncertainty.

3. Efficient Shipping Performance

The average delivery time across all customer orders was approximately 4 days, with only minor differences between regions. This indicates that the company's logistics network is operating efficiently.

4. Strong Seasonal Sales Pattern

Sales consistently increase during September, November, and December, indicating strong seasonal demand associated with festive shopping, promotional campaigns, and year-end purchasing.

5. Best Forecasting Model

Three forecasting models were compared:

Model	MAPE
SARIMA	21.05%
Prophet	21.86%
XGBoost	13.42%

XGBoost achieved the lowest forecasting error and is therefore recommended as the production forecasting model for future business planning.

Three-Month Sales Forecast

Although **XGBoost** produced the most accurate forecasts, the **SARIMA** model is presented below because it provides confidence ranges that help estimate the uncertainty surrounding future sales.

Forecast Month	Expected Sales	Expected Sales Range
Month 1	62,204	37,537 – 86,871
Month 2	88,918	64,251 – 113,585
Month 3	98,699	74,032 – 123,366

Sales are expected to remain strong over the next three months, with demand gradually increasing throughout the forecast period. While the forecast provides the most likely sales values, actual sales may vary within the estimated ranges because of factors such as customer demand, promotional campaigns, market conditions, seasonal shopping behaviour, and supply chain disruptions. These ranges should be used during budgeting, purchasing, and inventory planning to minimize both stock shortages and excess inventory.

Top Three Sales Anomalies and Their Likely Causes

Date	Observation	Likely Business Cause
22 March 2015	Exceptionally high weekly sales	Major promotional campaign, seasonal demand, or large corporate purchases.
22 February 2015	Extremely low weekly sales	Possible inventory shortages, reduced customer demand, or operational disruptions.
17 December 2017	Significant sales spike	Increased customer purchases during Christmas and year-end holiday shopping.

These unusual sales events highlight the importance of preparing inventory and marketing strategies for promotional periods while monitoring unexpected demand declines.

Product Demand Segmentation and Recommended Stocking Strategy

The forecasting analysis identified different demand patterns across product categories and regions.

1. Office Supplies

Office Supplies are expected to maintain the most consistent and stable demand over the next three months.

Recommended Strategy

- Maintain higher inventory levels.
- Ensure continuous product availability.
- Reduce the risk of stock shortages by prioritizing replenishment.

2. Technology

Technology products generate the highest revenue but experience moderate fluctuations in demand.

Recommended Strategy

- Maintain sufficient inventory levels.
- Monitor demand closely during promotional campaigns and festive seasons.
- Replenish inventory frequently to prevent stockouts.

3. Furniture

Furniture demand is comparatively more variable and less predictable.

Recommended Strategy

- Maintain moderate inventory levels.
- Restock based on actual sales trends rather than large advance purchases.
- Avoid overstocking high-value products.

4. Regional Inventory Strategy

The **East region** is forecasted to experience the strongest future sales growth, while the **Central region** remains the most stable market.

Recommended Strategy

- Allocate additional inventory to the East region to support expected demand growth.
- Continue using the Central region as a benchmark for inventory planning due to its predictable sales pattern.

Business Recommendations

1. Increase Inventory Before Peak Sales Months

Historical data shows that sales consistently rise during September, November, and December. Increasing inventory before these months will reduce stock shortages and improve customer satisfaction during peak demand periods.

2. Prioritize Office Supplies and the East Region

Forecast results indicate that Office Supplies will experience stable demand while the East region is expected to record the strongest future growth. Prioritizing inventory allocation in these areas is likely to improve sales performance and operational efficiency.

3. Deploy XGBoost as the Company's Forecasting Model

Among all forecasting models tested, XGBoost achieved the highest prediction accuracy with the lowest forecasting error (MAPE = 13.42%). Using this model for future forecasting will improve purchasing decisions, inventory planning, and financial forecasting.

Risk and Limitation

The forecasting system is based primarily on historical sales data. Unexpected factors such as economic downturns, competitor actions, supply chain disruptions, changes in customer preferences, or new market trends may influence future sales differently from past patterns. Therefore, forecasts should be reviewed regularly and combined with business expertise before making major operational or financial decisions.

Conclusion

The Sales Forecasting and Demand Intelligence System provides valuable insights into customer demand, seasonal buying patterns, inventory requirements, and future sales performance. The analysis demonstrates that Technology remains the largest revenue-generating category, Office Supplies is expected to deliver the most stable future demand, and the East region is likely to experience the strongest sales growth. While the SARIMA model provides useful confidence ranges for business planning, XGBoost achieved the highest forecasting accuracy and is recommended for production deployment. By implementing the recommendations outlined in this report, the company can improve inventory management, optimize stock allocation, reduce operational costs, and make more informed financial and supply chain decisions.